

SMART INDIA HACKATHON 2026

TITLE PAGE



SMART INDIA
HACKATHON
2026

Problem Statement ID – PS-XXXXX

Problem Statement Title – Disaster Evacuation System

Theme – Disaster Management

PS Category – Software

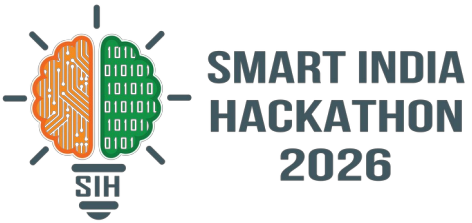
Team ID – TXXXXX

Team Name – Your Team Name



Your
Team
Name

RAKSHA-NET



Proposed Solution — Unified Alert, Rescue & Reunification Platform

Multi-hazard alerts: NDMA & IMD warnings for earthquake, flood, cyclone, fire and landslide via app, SMS and Cell Broadcast.

One-tap SOS dispatch: Live GPS shared instantly with the nearest police, ambulance, fire or NDRF unit.

“Find My Missing”: Photo + last-known-location registry, crowd-sourced sightings, AI facial matching and shelter check-in/out.

AI-assisted safe routing: Dynamic evacuation paths to relief camps avoid flooded, blocked or collapsed routes using live reports.

HOW IT ADDRESSES THE PROBLEM

Closes the gap between government warning systems and citizens on the ground.

Cuts emergency response time and reunites separated families faster, reducing casualties and panic.

INNOVATION & UNIQUENESS

Combines early warning, SOS dispatch and missing-person tracing in one platform.

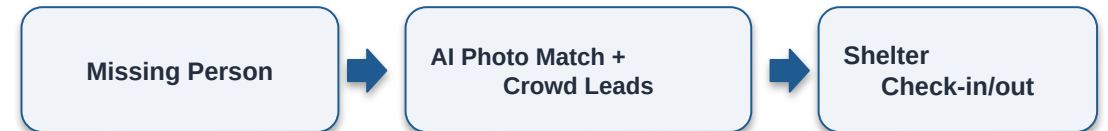
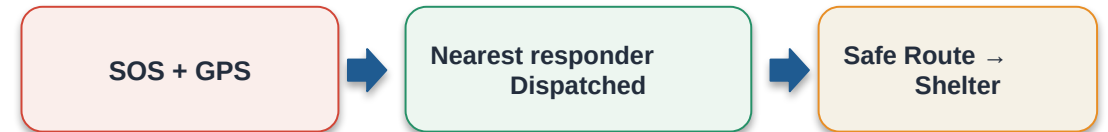
Works during network outages via peer-to-peer Bluetooth/Wi-Fi Direct mesh relay.

Crowd-verified missing-person leads instead of unverified social-media chaos.

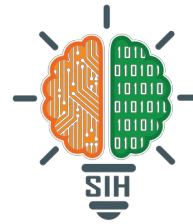
TECHNOLOGIES TO BE USED

Citizen App	Flutter / React Native
Control-room Dashboard	React.js + Node.js
Backend	Django REST / Node.js microservices
Database	PostgreSQL + PostGIS, Firebase RTDB, Redis
Alerts	NDMA SACHET, IMD, Cell Broadcast, FCM, Twilio
Maps / Routing	OpenStreetMap + Mapbox, A* dynamic routing
AI / ML	OpenCV photo matching, NLP distress parsing
Offline Layer	Bluetooth Mesh / Wi-Fi Direct relay
Infrastructure	AWS/GCP, Docker, Kubernetes

METHODOLOGY / PROCESS FLOW



All three streams converge at the NDRF / SDMA Control-Room Dashboard for real-time coordination.



ANALYSIS OF THE FEASIBILITY OF THE IDEA

Built on NDMA SACHET, IMD and Cell Broadcast — no need to build hazard-prediction models from scratch.

Modular rollout: Alerts first → SOS dispatch second → missing-person module third.

Google Person Finder and Japan's J-Alert provide real-world precedents for similar systems.

MODULAR ROLLOUT

1 Alerts

2 SOS Dispatch

3 Missing-Person

POTENTIAL CHALLENGES & STRATEGIES

Network outage during disasters → Offline Bluetooth/Wi-Fi mesh relay + SMS fallback.

False alerts causing panic → Verified government feeds + admin moderation.

Low adoption in rural/elderly population → Multilingual voice alerts + Panchayat/NGO onboarding.

Traffic spikes during actual disasters → Cloud auto-scaling + CDN caching.

Privacy of missing-person data → Encryption, consent-based sharing + auto-deletion after case closure.

POTENTIAL IMPACT ON THE TARGET AUDIENCE

Citizens

Earlier, more actionable warnings — more time to move to safety.

Emergency Services

Faster access to people during the critical “golden hour”.

Families

Faster reunification of separated family members, reducing trauma.

NDMA / SDMA / NDRF

Real-time ground picture for better resource deployment.

BENEFITS

SOCIAL

Reduced panic; inclusive access for elderly/disabled; peace of mind for families.

ECONOMIC

Lower disaster-relief costs via efficient allocation; less property/business loss.

ENVIRONMENTAL Better coordination limits secondary disasters — fire spread and stampedes.

GOVERNANCE Structured incident data supports better infrastructure and policy planning.

Your
Team
Name

RESEARCH AND REFERENCES



**SMART INDIA
HACKATHON
2026**

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

- | | | |
|---|---|--|
| 1 | National Disaster Management Authority (NDMA) | ndma.gov.in |
| 2 | SACHET Common Alerting Protocol | sachet.ndma.gov.in |
| 3 | India Meteorological Department | mausam.imd.gov.in |
| 4 | Google Person Finder | Crowd-sourced missing-person case study used in past disasters worldwide |
| 5 | Department of Telecommunications — Cell Broadcast Alert System (CBAS) | Documentation |

RAKSHA-NET — Smart India Hackathon